Optimized Data Profiling Report.docx

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1. Data Profiling (Power BI)

Data profiling was conducted in Power Query to assess the quality, structure, and consistency of the Movies dataset prior to transformation and analysis. The following profiling tools were applied:

- Column Quality was used to evaluate data validity by identifying valid, missing, and error values across all attributes.
- Column Distribution was used to examine frequency patterns and identify inconsistencies in categorical variables.
- Column Profile was used to generate statistical summaries, including minimum, maximum, and distribution insights for each column.

Key Findings from Profiling

Data profiling identified several data quality and structural issues:

- Missing values were detected in selected fields and were resolved through removal or standardisation to ensure dataset completeness and reliability.
- Minor inconsistencies were observed in categorical fields such as Language and Country, which were standardised for consistency across the dataset.
- Validation of Total Views confirmed no significant outliers or structural anomalies.
- Overall, the dataset demonstrated a high proportion of valid and usable records following initial assessment.

2. Data Quality Issues Identified and Resolved

The following issues were identified during profiling and addressed during the cleaning and transformation process:

1. A derived Year column was created by extracting values from the Title field to support time-based analysis.
2. The invalid value "non_def" was corrected and replaced with the appropriate year value (1995).
3. The incorrectly labelled film title "Untitled" was corrected to *The Phantom of the Opera*.
4. Genre delimiter inconsistencies were resolved by standardising separators from "–" to "|", ensuring consistent multi-value formatting (e.g., Film-Noir → Film|Noir).
5. Spelling inconsistencies in the Drama genre were corrected by standardising all variations (e.g., "Dramatic", "Dramma") to DRAMA.

3. Data Cleaning and Transformation Process

The data cleaning process followed a structured pipeline approach:
Structure → Cleaning → Transformation → Validation
This ensured a systematic and efficient workflow that minimised redundancy and maintained data integrity throughout the process.

3.1 Data Source Preparation

- Data was imported into Power BI with automatic header promotion.
- No structural adjustments were required at ingestion stage.

3.2 Data Type Validation

Appropriate data types were assigned to ensure analytical accuracy:

- MovieID → Whole Number
- Title → Text
- Genres → Text
- Language → Text
- Country → Text
- Total Views → Whole Number

3.3 Data Cleaning

- Blank rows were removed from key fields (MovieID, Title).
- Error values were identified and removed from MovieID and Total Views.
- Duplicate records were removed based on MovieID to ensure unique movie identification.

3.4 Data Transformation

- The Year field was extracted from the Title column using a right-most delimiter split.
- Extracted values were cleaned and converted into Whole Number format.
- Columns were renamed appropriately to improve clarity and consistency.

3.5 Text Standardisation

- Trim and Clean functions were applied to Title, Genres, Language, and Country fields.
- Genres were converted to uppercase to ensure consistent classification standards.

3.6 Value Standardisation

- Minimal value replacement was performed.
- Only abnormal values (e.g., “non_def”) were replaced with null values to maintain data integrity.

3.7 Final Validation

Final checks ensured consistency across key fields:

- Year → Whole Number
- Total Views → Whole Number
- No structural or formatting inconsistencies detected in cleaned dataset

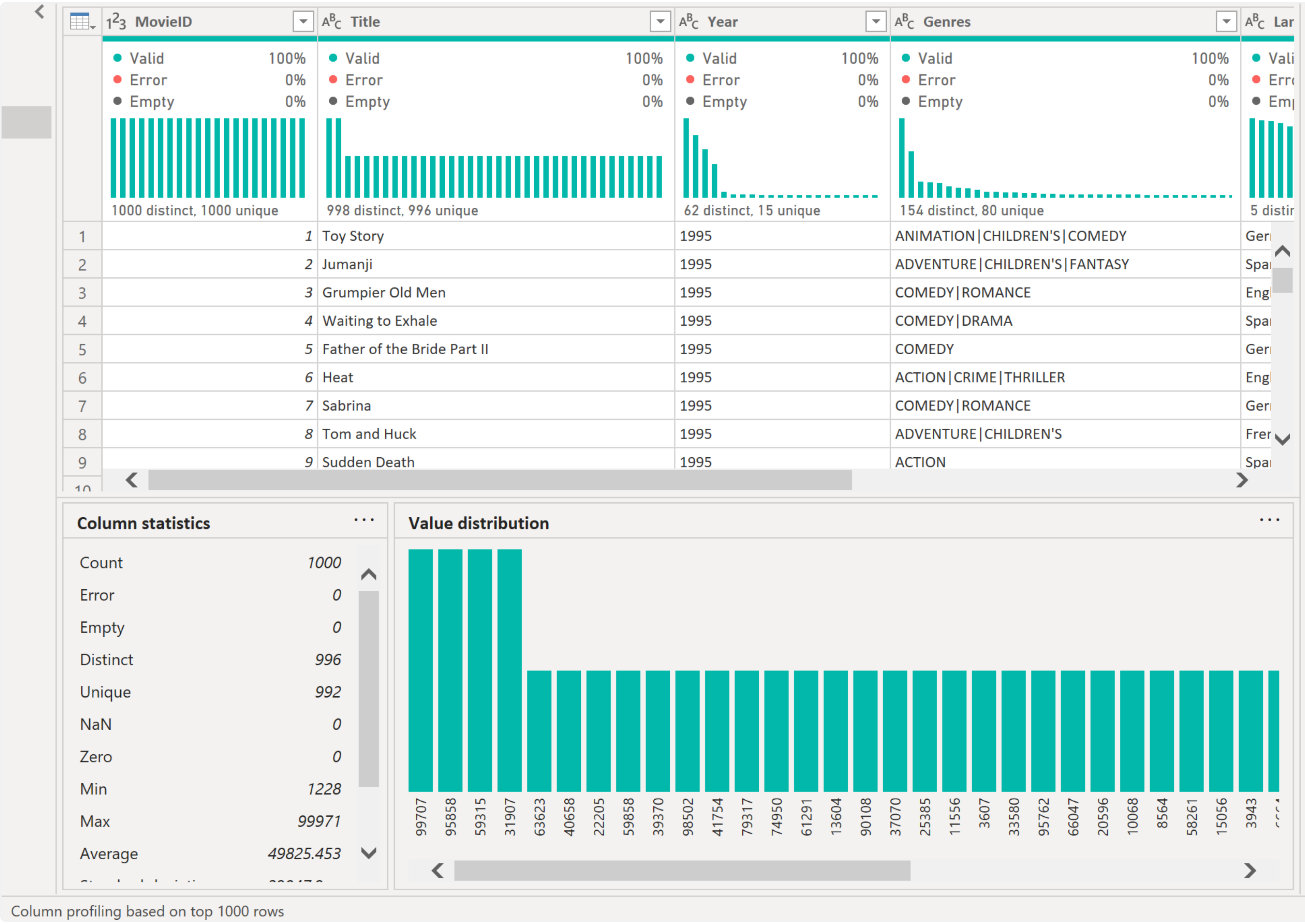
4. Final Outcome

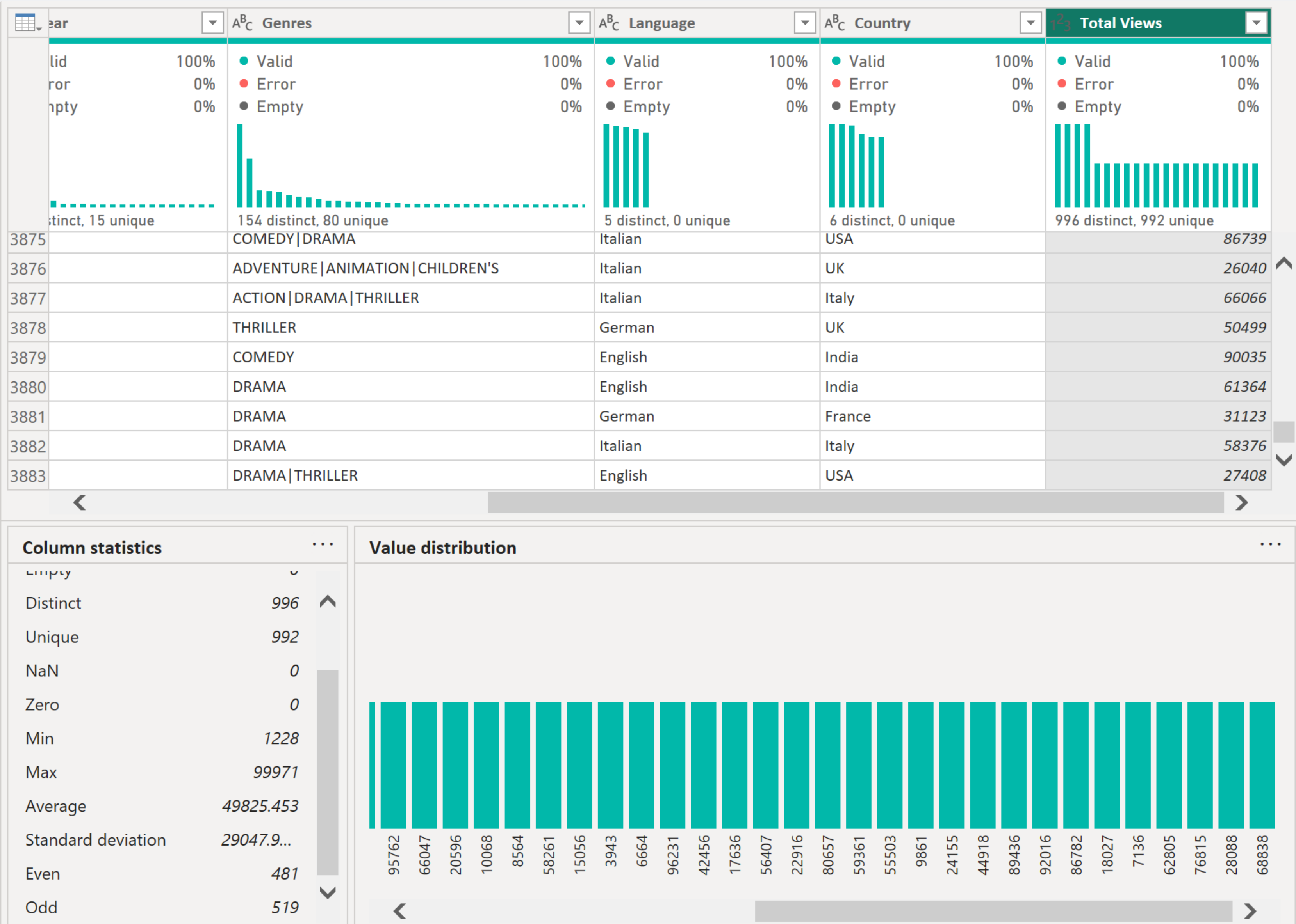
- The dataset was successfully transformed into a clean, standardised, and analysis-ready format.
- Data quality and structural consistency were significantly improved across all attributes.
- Multi-value attributes (Genres) were normalised into a relational structure suitable for analytical modelling.

- The dataset is fully compatible with a Star Schema design for Power BI reporting and dashboard development.

5. Final Conclusion

The dataset was successfully transformed from a semi-structured raw dataset into a fully cleaned and standardised analytical dataset. The structured profiling and transformation process ensured high data integrity, improved consistency, and enabled reliable downstream analysis and visualisation in Power BI.





ear
Genres
Language
Country
Total Views
lid 100%
ror 0%
npty 0%
Valid 100%
Error 0%
Empty 0%
Valid 100%
Error 0%
Empty 0%
Valid 100%
Error 0%
Empty 0%
Valid 100%
Error 0%
Empty 0%
stinct, 15 unique
154 distinct, 80 unique
5 distinct, 0 unique
6 distinct, 0 unique
996 distinct, 992 unique
3875 COMEDY|DRAMA Italian USA 86739
3876 ADVENTURE|ANIMATION|CHILDREN'S Italian UK 26040
3877 ACTION|DRAMA|THRILLER Italian Italy 66066
3878 THRILLER German UK 50499
3879 COMEDY English India 90035
3880 DRAMA English India 61364
3881 DRAMA German France 31123
3882 DRAMA Italian Italy 58376
3883 DRAMA|THRILLER English USA 27408
Column statistics
Distinct 996
Unique 992
NaN 0
Zero 0
Min 1228
Max 99971
Average 49825.453
Standard deviation 29047.9...
Even 481
Odd 519
Value distribution
95762
66047
20596
10068
8564
58261
15056
3943
6664
96231
42456
17636
56407
22916
80657
59361
55503
9861
24155
44918
89436
92016
86782
18027
7136
62805
76815
28088
68838